Bandit Algorithms

Sri Architha, Dattabhasvant G

Introduction

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Multi-Armed Bandit Problem

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$$\mathcal{E} = \times_{a \in \mathcal{A}} \mathcal{M}_a$$

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- $\mathcal{H}_t = \{(a_1, x_1, ..., a_t, x_t) : a_i \in \mathcal{A}, x_i \in \mathbb{R}\}$

Multi-Armed Bandit Problem

$\pi_t : \mathcal{A} \times \mathcal{H}_{t-1} \rightarrow [0, 1]$, representing the probability of pulling and arm a given the history.
A **policy** π is the sequence, $(\pi_t)_{t=1}^{n}$ Note that,

$$p_{\nu\pi}(a_1, x_1, ..., a_n, x_n) = \prod_{t=1}^{n} \pi_t(a_t|a_1, x_1, ...a_{t-1}, x_{t-1}) P_{a_t}(x_t)$$

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- Environment samples $X_t \in \mathbb{R}$ from P_{A_t}, and reveals the reward.
- The learner's objective: Maximize the expected total reward from the environment, $\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^{n} X_t\right]$
- Equivalently, we define a quantity called Regret that we minimize.

Regret

- Given a policy π and stochastic bandit ν, **Regret** over n time steps is defined as

$$R_n(\pi, \nu) = \sum_{t=1}^{n} \mathbb{E}[\mu^*(\nu) - X_t]$$

where $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$, the expected reward of the optimal arm.

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- **Regret Decomposition**

$$R_n = \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}\left[T_a(n)\right]$$

Subgaussian Random Variables

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$$\mathbb{E}[e^{\lambda X}] \leq e^{\lambda^2\sigma^2/2}$$

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- **Theorem.** If X is σ-subgaussian, then for any $\epsilon \geq 0$,

$$\mathbb{P}(X \geq \epsilon) \leq e^{-\epsilon^2/2\sigma^2} \qquad \mathbb{P}\left(X \leq \sqrt{2\sigma^2 \log(1/\delta)}\right) \leq \delta$$

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- Let X_1 and X_2 be σ_1 and σ_2 subgaussian respectively and independent. Then,
  1. If $\mathbb{E}[X_1] = 0$ then $Var(X_1) \leq \sigma_1^2$
  2. cX_1 is $(|c|\sigma_1)$-subgaussian for all $c \in \mathbb{R}$
  3. $X_1 + X_2$ is $(\sqrt{\sigma_1^2 + \sigma_2^2})$-subgaussian

Hoeffding's Inequality

Hoeffding's Inequality If X_i's are independent σ-subgaussian random variables with mean μ and $\hat{\mu}(n) = \frac{1}{n}\sum_{t=1}^{n} X_t$. Then for any $\epsilon \geq 0$,

$$\mathbb{P}(\hat{\mu}(n) \leq \mu + \epsilon) \leq e^{-\frac{n\epsilon^2}{2\sigma^2}} \qquad \mathbb{P}(\hat{\mu}(n) \geq \mu - \epsilon) \leq e^{-\frac{n\epsilon^2}{2\sigma^2}}$$

Basic Algorithms

Explore-Then-Commit

Algorithm 1 ETC Algorithm

1: Input **m**

2: For each round, t, choose;

$$A_t = \begin{cases} (t \mod k) + 1, \text{ if } t \leq mk \\ \text{argmax}_{a \in \mathcal{A}} \hat{\mu}_a(mk), \text{ otherwise} \end{cases}$$

Theorem

$$R_n \leq m \sum_{a \in \mathcal{A}} \Delta_a + (n - mk) \sum_{a \in \mathcal{A}} \Delta_a \exp\left(\frac{-m\Delta_a^2}{4}\right)$$

for $1 \leq m \leq n/k$

Upper Confidence Bound

Algorithm 2 UCB(δ)

1: Input k **and** δ
2: **for** $t \in 1, \ldots, n$ **do**
3: Choose $A_t = \text{argmax}_a \left(\hat{\mu}_a(t) + \sqrt{\frac{2\log(1/\delta)}{T_a(t-1)}} \right)$
4: Observe the reward X_t and update UCB's
5: **end for**

Theorem

For any horizon n, if $\delta = 1/n^2$ the UCB(δ) algorithm suffers regret bounded above by

$$R_n \leq 3 \sum_{a \in \mathcal{A}} \Delta_a + \sum_{a:\Delta_a > 0} \frac{16 \log n}{\Delta_a} \tag{1}$$

Regret Analysis for UCB

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Define the 'good' event for arm a,

$$G_a := \left\{ \mu_1 < \min_{t \in [n]} \mathrm{UCB}_1(t,\delta) \right\} \bigcap \left\{ \hat{\mu}_{a,u_a} + \sqrt{\frac{2\log(1/\delta)}{u_a}} < \mu_1 \right\}$$

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To bound the Regret,

$$\mathbb{E}[T_a(n)] = \mathbb{E}[\mathbb{I}_{G_a} T_a(n)] + \mathbb{E}[\mathbb{I}_{G_a^c} T_a(n)] \le T_a(n) + n\mathbb{P}(G_a^c)$$

Regret Analysis for UCB (contd.)

- Suppose G_a occurred but $T_a(n) > u_a$. So when, $T_a(t-1) = u_a$ and $A_t = a$, $UCB_a(t-1,\delta) < \mu_1 < UCB_1(t-1,\delta)$. But then $A_t = 1$.

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- $\mathbb{P}(E_1^c) \leq \sum_{s=1}^{n} \mathbb{P}\left(\mu_1 \geq \hat{\mu}_{1,s} + \sqrt{\frac{2\log(1/\delta)}{s}}\right) \leq n\delta$

Regret Analysis for UCB (contd.)

Bandit Algorithms

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Basic Algorithms

Regret Lower Bounds

Optimal Algorithms

Pure Exploration

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- Choose u_a large enough so that $(1-c)\Delta_a \geq \sqrt{\frac{2\log(1/\delta)}{u_a}}$,

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- Choose u_a large enough so that $(1-c)\Delta_a \geq \sqrt{\frac{2\log(1/\delta)}{u_a}}$, then

$$\mathbb{P}(E_2^c) = \mathbb{P}\left(\hat{\mu}_{a,u_a} - \mu_a \geq \Delta_a - \sqrt{2\log(1/\delta)/u_a}\right)$$

$$\leq \mathbb{P}\left(\hat{\mu}_{a,u_a} - \mu_a \geq c\Delta_a\right) \leq e^{-\frac{u_a c^2 \Delta_a^2}{2}}$$

- $\mathbb{E}\left[T_a(n)\right] \leq u_a + n\left(n\delta + \exp\left(-\frac{u_a c^2\Delta_a^2}{2}\right)\right)$

Regret Analysis for UCB (contd.)

Theorem

For any $\nu \in \mathcal{E}_{SG}^{k}(1)$, if $\delta = 1/n^2$, the regret of UCB is bounded by,

$$R_n \leq 8\sqrt{kn \log n} + 3\sum_{a \in \mathcal{A}} \Delta_a$$

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- For $\Delta > 0$,

$$\begin{aligned} R_n &= \sum_{a:\Delta_a < \Delta} \Delta_a \mathbb{E}\left[T_a(n)\right] + \sum_{a:\Delta_a \geq \Delta} \Delta_a \mathbb{E}\left[T_a(n)\right] \\ &\leq n\Delta + \frac{16k \log(n)}{\Delta} + 3 \sum_{a \in \mathcal{A}} \Delta_a \end{aligned}$$

- Choose $\Delta = \sqrt{(16k \log n)/n}$

Regret Lower Bounds

Instance Dependent Lower Bounds

- A policy is called **consistent** if for every $\nu \in \mathcal{E}$ and $p > 0$,

$$\lim_{n \to \infty} \frac{R_n(\nu, \pi)}{n^p} = 0$$

- Suppose we have a collection of distributions with finite means, $\mathcal{M}$. For $P \in \mathcal{M}$ and $\mu_0 > \mu(P)$ define

$$KL_{inf}(P, \mu_0) = \inf_{P' \in \mathcal{M}} \{KL(P||P') : \mu(P') > \mu_0\}$$

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Theorem

For $\pi \in \Pi_{cons}(\mathcal{E})$ and $\nu \in \mathcal{E}$,

$$\liminf_{n\to\infty} \frac{R_n(\nu,\pi)}{\log n} \geq \sum_{a:\Delta_a>0} \frac{\Delta_a}{KL_{inf}(P_a,\mu^*)}$$

Minimax Lower Bounds

- For a policy π, the **Worst-case Regret** over an environment class is

$$R_n(\pi, \mathcal{E}) = \sup_{\nu \in \mathcal{E}} R_n(\pi, \nu)$$

Minimax Lower Bounds

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- For *the best possible performance in the worst environment*, define the **Minimax Regret**,

$$R_n^*(\mathcal{E}) = \inf_{\pi \in \Pi} \sup_{\nu \in \mathcal{E}} R_n(\pi, \nu)$$

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- For the environment class of k-armed subgaussian bandits, $R_n^*(\mathcal{E}_{SG}^k) = \Theta(\sqrt{kn})$

Optimal Algorithms

Optimality

- A policy is **asymptotically optimal** if

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- A policy is **minimax optimal** if

$$R_n(\pi,\nu) = R_n^*(\mathcal{E})$$

Minimax Optimal Strategy in Stochastic Case

Algorithm 3 MOSS

Choose each arm once. Subsequently choose

$$A_t = \text{argmax}_{a \in \mathcal{A}} \left(\hat{\mu}_a(t-1) + \sqrt{\frac{4 \log^+(n/kT_a(t-1))}{T_a(t-1)}} \right)$$

- MOSS can perform significantly worse than UCB in specific problem instances
- It is useful when suboptimality gaps, Δ_a's are unknown or small

Asymptotically Optimal UCB

Algorithm 4 UCB_{LR}

Choose each arm once. Subsequently choose

$$A_t = \text{argmax}_{a \in \mathcal{A}} \left(\hat{\mu}_a(t-1) + \sqrt{\frac{2 \log f(t)}{T_a(t-1)}} \right)$$

where $f(t) = 1 + t \log^2 t$.

- For 1-Subgaussian bandits,

$$\limsup_{n \to \infty} \frac{R_n}{\log(n)} \leq \sum_{a:\Delta_a > 0} \frac{2}{\Delta_a}$$

- For Gaussian bandits, $KL_{inf}(P, \mu^*) = (\mu - \mu^*)^2 / 2\sigma^2$, so the bound matches exactly

KL-UCB

Algorithm 5 KL-UCB (Bernoulli)

Choose each arm once. Subsequently choose

$$A_t = \text{argmax}_{a \in \mathcal{A}} \max \left\{ \tilde{\mu} : d(\hat{\mu}_a(t-1), \tilde{\mu}) \leq \frac{\log f(t)}{T_a(t-1)} \right\}$$

where $f(t) = 1 + t \log^2 t$

- To generalize to other single parameter distributions, replace $d(\cdot, \cdot)$ with $KL(P_{\theta(x)} || P_{\theta(q)})$.
- Asymptotically optimal as well; satisfies the Lai-Robbins Lower Bound

Pure Exploration

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$$\mathbb{P}(\mu^* - \mu_{a_{output}} \leq \varepsilon) \geq 1 - \delta$$

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- A (ε, δ)-**PAC algorithm** outputs an ε-optimal arm with probability $\geq 1 - \delta$.

$$\mathbb{P}(\mu^* - \mu_{a_{output}} \leq \varepsilon) \geq 1 - \delta$$

$$\begin{array}{ccc} & \text{Settings} & \\ \swarrow & & \searrow \\ \text{Fixed Confidence} & & \text{Fixed Budget} \end{array}$$

Baseline: Naive Sampling

Algorithm 6 Naive(ε, δ) Algorithm

Sample each arm $\ell = O\left(\frac{1}{\varepsilon^2}\log\frac{k}{\delta}\right)$ times and return empirical best.

Sample complexity: $O\left(\frac{k}{\varepsilon^2}\log\frac{k}{\delta}\right)$.

Successive Elimination

Algorithm 7 Successive Elimination

1: Set $S = A$
2: Set for every arm a, $\hat{\mu}_a^1 = 0$
3: Sample every arm $a \in S$ once and let $\hat{\mu}_a^t$ be the average reward of arm a by time t.
4: Let $\hat{\mu}_{max}^t = \max_{a \in S}\{\hat{\mu}_a^t\}$ and $\alpha_t = \sqrt{\frac{\log(2kt^2/\delta)}{t}}$
5: For every arm $a \in S$ such that $\hat{\mu}_{max}^t - \hat{\mu}_a^t \geq 2\alpha_t$, set $S = S \setminus \{a\}$
6: $t = t + 1$
7: if $|S| > 1$, go to step 3; else, output S

Successive Elimination Complexity

Theorem (Karnin et al. 2013)

Successive Elimination is $(0, \delta)$-PAC and with probability at least $1 - \delta$ the sample complexity is bounded by:

$$O\left(\sum_{a \in \mathcal{A}} \frac{1}{\Delta_a^2} \log \frac{k}{\delta \Delta_a}\right)$$

Median Elimination

Algorithm 8 Median Elimination(ϵ, δ)

1: Set $S_1 = A$.
2: $\epsilon_1 = \epsilon/4$, $\delta_1 = \delta/2$, $r = 1$.
3: Sample every arm $a \in S_r$ for $\frac{4}{\epsilon_r^2} \ln(3/\delta_r)$ times
4: Find the median of $\{\hat{\mu}_a^r\}_{a \in S_r}$, denoted by m_r.
5: $S_{r+1} = S_r \setminus \{a : \hat{\mu}_a^r < m_r\}$.
6: if $|S_r| = 1$, output S_r
7: Else $\epsilon_{r+1} = \frac{3}{4}\epsilon_r$, $\delta_{r+1} = \delta_r/2$, $r = r + 1$ and go to step 3.

Median Elimination Complexity

Theorem

Median Elimination is (ε, δ)-PAC with sample complexity:

$$O\left(\frac{k}{\varepsilon^2}\log\frac{1}{\delta}\right).$$

Exponential-Gap Elimination (EGE)

Algorithm 9 Exponential-Gap Elimination

1: Input confidence $\delta > 0$
2: initialize $S_1 \leftarrow [k]$, $r \leftarrow 1$,
3: let $\varepsilon_r = 2^{-r}/4$ and $\delta_r = \delta/(50r^3)$
4: sample each arm $a \in S_r$ for $t_r = (2/\varepsilon_r^2)\ln(2/\delta_r)$ times, and let $\hat{\mu}_a^r$ be the average reward
5: invoke $a_r \leftarrow \textsc{MedianElimination}(S_r, \varepsilon_r/2, \delta_r)$ and let $\hat{\mu}_*^r = \hat{\mu}_{a_r}^r$
6: set $S_{r+1} \leftarrow S_r \setminus \{a \in S_r : \hat{\mu}_a^r < \hat{\mu}_*^r - \varepsilon_r\}$
7: if $|S_r| = 1$, output S_r, else update $r \leftarrow r + 1$ and go to step 3.

EGE Complexity

Theorem (Karnin et al. 2013)

Exponential-Gap Elimination is $(0, \delta)$-PAC with sample complexity:

$$O\left(\sum_{a \neq a^*} \frac{1}{\Delta_a^2} \log\left(\frac{1}{\delta} \log \frac{1}{\Delta_a}\right)\right).$$

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